R16

Code No: 131AK

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year I Semester Examinations, May - 2018
BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Define dependent sources. [2]
b) The essential component of a toaster is an electrical element (a resistor) that converts electrical energy to heat energy. How much current is drawn by a toaster with resistance 15Ω at 110 V? [3]
c) What is the condition for maximum power transfer in ac circuits? [2]
d) Derive the expression for resonant frequency. [3]
e) Define reverse break down voltage of a diode. [2]
f) Derive an expression for the average output voltage of a full wave rectifier. [3]
g) Draw the high frequency model of a transistor. [2]
h) In a BJT, the emitter current is 12 mA and the emitter current is 1.02 times the collector current. Find the base current. [3]
i) Why FET is called unipolar device? [2]
j) What are the special features of FET? [3]

PART-B

(50 Marks)

2.a) Differentiate between active and passive elements.
b) Obtain the equivalent resistance for the circuit in figure 1 and use it to find current *i*. [4+6]

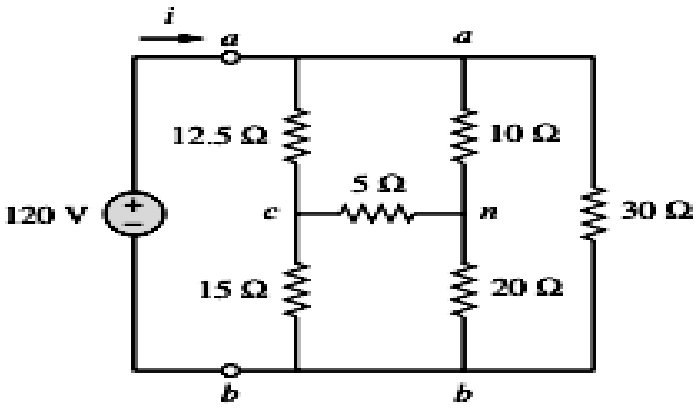


Figure: 1
OR

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3.a) Define impedance, susceptance and admittance with examples.

b) Using mesh analysis, find I_o in the circuit shown in figure 2. [4+6]

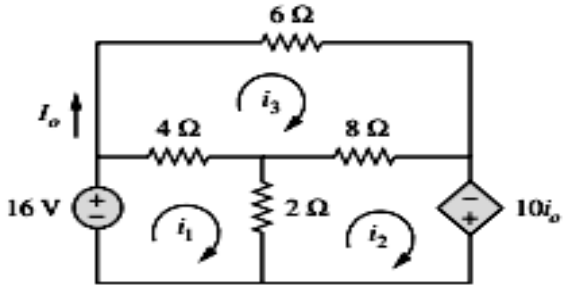


Figure: 2

4.a) The Q factor of a RLC series circuit is 5 at a resonance frequency of 1 KHz. Assuming the power dissipation of 250W, when the current drawn is 1A, find the circuit parameters.

b) Using superposition, find V_o in the circuit shown in figure 3. [5+5]

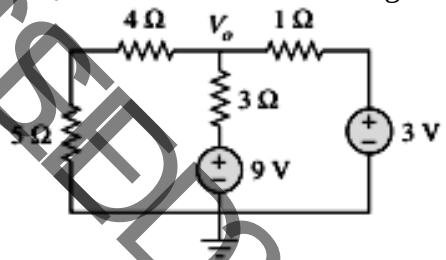


Figure: 3

OR

5.a) A 5 μF condenser is connected in series with a coil having inductance of 50mH. Determine the frequency of resonance, the resistance of the coil if a 50V source causing a current of 10mA is connected. What is the Q factor of the coil?

b) Use Norton's theorem to find V_o in the circuit shown in figure 4. [4+6]

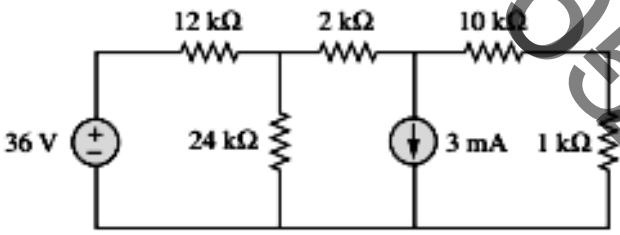


Figure: 4

6.a) Draw the V – I characteristics of a silicon diode, write an expression for diode current and discuss how cut in voltage changes with temperature.

b) Draw the circuits of a full wave rectifier using 2-diodes and 4-diodes. Discuss the relative merits and demerits. [5+5]

OR

7.a) Define diffusion capacitance in a P-N junction diode and discuss its dependence on diode biasing.

b) Derive expressions for ripple factor and efficiency of rectification for a full wave rectifier. [5+5]

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8.a) Derive expressions for voltage gain, input-resistance and current gain for an emitter follower.

b) In the Silicon Transistor circuit shown in figure 5, find the maximum base resistance R_B to drive the Transistor into saturation. [5+5]

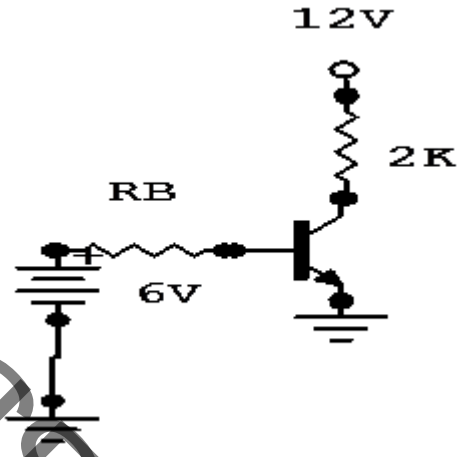


Figure: 5

OR

9.a) Discuss the need for biasing a transistor. Briefly explain about self biasing.

b) Explain how h-parameters can be determined from the static input and output characteristics. [5+5]

10.a) Discuss the characteristic differences between a BJT and a FET. Draw a diagram depicting the structure of a N-channel FET and identify the various terminals and the biasing voltages.

b) Explain how the pinch off voltage can be modified without changing the physical structure of a JFET. [5+5]

OR

11.a) Explain the operation of a FET with a neat diagram and V-I characteristics.

b) An N-channel JFET has a pinch voltage of -4.5V and I_{DSS} = 9mA at what value of V_{GS} will I_{DS} equal to 3mA? What is its g_m at this I_{DS}. [5+5]

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